

Submission LINALG1-FINAL-SUB08

Question LINALG1-FINAL-Q01

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visible work:

$$\det = 6(6) - 4(4) = 20 \dots ?$$

last line is hard to read; maybe $x=1$, $y=2$

Question LINALG1-FINAL-Q02

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visible work:

$$\det(A) = 2 \cdot 6 - 2 \cdot 2 = 8 \dots ?$$

last line is hard to read; maybe $\det=8$; invertible

Question LINALG1-FINAL-Q03

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visible work:

$$T(1, -1) = (1 - 2, 3 + 1) = (-1, 4) \dots ?$$

last line is hard to read; maybe $T(1, -1) = (-1, 4)$; matrix $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$; linear

Question LINALG1-FINAL-Q04

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visible work:

$$\text{Set } c \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 0 \\ 1 \end{bmatrix} = (0, 0) \dots ?$$

last line is hard to read; maybe only trivial combination; $\{(1, 0), (0, 1)\}$ is linearly independent

Question LINALG1-FINAL-Q05

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visible work:

$\text{Null}(A)$ is the set of vectors sent to the zero vector.... ?

last line is hard to read; maybe the vectors mapped to zero; exactly the solution space of $Ax=0$

Question LINALG1-FINAL-Q06

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visible work:

$$\det = 6(6) - 1(4) = 32 \dots ?$$

last line is hard to read; maybe $x=1$, $y=2$

Question LINALG1-FINAL-Q07

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visible work:

$$\det(A) = 4 \cdot 4 - 2 \cdot 2 = 12 \dots ?$$

last line is hard to read; maybe $\det=12$; invertible

Question LINALG1-FINAL-Q08

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visible work:

$$T(1, -1) = (1 - 2, 3 + 1) = (-1, 4) \dots ?$$

last line is hard to read; maybe $T(1, -1) = (-1, 4)$; matrix $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$; linear

Question LINALG1-FINAL-Q09

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visible work:

$$\text{Set } c_1(1, 0) + c_2(0, 1) = (0, 0) \dots ?$$

last line is hard to read; maybe only trivial combination; $\{(1, 0), (0, 1)\}$ is linearly independent

Question LINALG1-FINAL-Q10

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visible work:

$\text{Null}(A)$ is the set of vectors sent to the zero vector.... ?

last line is hard to read; maybe the vectors mapped to zero; exactly the solution space of $Ax=0$